1. The Structure of a Web Page (HTML)

HTML (HyperText Markup Language) defines the structure and content. An AL-standard document must follow the correct nesting order.

- **Essential Tags:**
  - `<!DOCTYPE html>`: Defines the document type.
  - `<html>`: The root element.
  - `<head>`: Contains metadata (e.g., `<title>`, `<meta>`, `<link>` for CSS).
  - `<body>`: Contains the visible content of the page.
- **Common Elements:**
  - **Text:** `<h1>` to `<h6>` (headings), `<p>` (paragraph), `<br>` (line break).
  - **Formatting:** `<b>` or `<strong>` (bold), `<i>` or `<em>` (italic), `<u>` (underline).
  - **Lists:** `<ol>` (Ordered), `<ul>` (Unordered), and `<li>` (List Item).
  - **Media:** `<img src="url" alt="description">`.
  - **Hyperlinks:** `<a href="url">Click Here</a>`.

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2. Styling with CSS (Cascading Style Sheets)

CSS is used to separate **content** (HTML) from **presentation** (Design).

- **Three Ways to Apply CSS:**
  1. **Inline:** Applied directly inside an HTML tag using the `style` attribute.
  2. **Internal/Embedded:** Written inside the `<style>` tag within the `<head>` section.
  3. **External:** Written in a separate `.css` file and linked using `<link rel="stylesheet" href="style.css">`. (Recommended for AL projects).
- **The Box Model:** Every element is viewed as a box consisting of:
  - **Content:** The actual text or image.
  - **Padding:** Space between content and border.
  - **Border:** The line surrounding the padding.
  - **Margin:** Space outside the border, separating the element from others.

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3. Client-Side vs. Server-Side Scripting

The AL syllabus emphasizes the difference between where code is executed.

Feature	Client-Side Scripting	Server-Side Scripting
Execution	Runs on the user's web browser.	Runs on the web server.
Languages	JavaScript, VBScript.	PHP, Python, ASP.NET.
Purpose	Interactivity (animations, form validation).	Database access, user authentication.

Feature	Client-Side Scripting	Server-Side Scripting
Visibility	Source code is visible to the user.	Source code is hidden from the user.

Export to Sheets

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4. Dynamic Web Development (PHP & MySQL)

A **dynamic** website changes content based on user input or database records.

- **PHP (Hypertext Preprocessor):** A server-side language used to process forms and communicate with databases.
- **MySQL:** A Relational Database Management System (RDBMS) used to store website data (e.g., user profiles, product lists).
- **Connectivity:** PHP uses functions like `mysqli_connect()` to link the web page to the database, allowing for tasks like "User Login" or "Search."

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5. Website Publishing and Maintenance

- **Domain Name:** The unique address (URL) of the site (e.g., `www.example.lk`).
- **Web Hosting:** Storing the website files on a server that is connected to the internet 24/7.
- **FTP (File Transfer Protocol):** The protocol used to upload files from a local computer to the web server.
- **Maintenance:** Regularly updating content, fixing broken links, and ensuring security patches are applied.

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6. Key Concepts for the Exam

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- **Static vs. Dynamic:** Static pages stay the same for everyone; dynamic pages change (e.g., Facebook feed).
- **URL Structure:** Protocol (`http://`) + Domain (`www.site.com`) + Path (`/folder/page.html`).
- **Form Handling:** Using `<form method="POST" action="process.php">` to send data securely to the server.